Vertex Formula

1 Consider the parabola of the form $y = ax^2 + bx + c$, where $a \neq 0$.
Complete the following statement:
The x-coordinate of the vertex of the parabola occurs at $x = \square$. The y-coordinate of the vertex is found by substituting $x = \square$ into the parabola's equation and evaluating the function at this value of x.

2 What does the line $x = \dfrac{-b}{2a}$ represent on the parabola defined by the equation $y = ax^2 + bx + c$ $(a \neq 0)$?

3 Consider the equation $y = 6x - x^2$.

- **a** Find the x-intercepts of the quadratic function.
- **b** Find the coordinates of the turning point.

4 Consider the parabola $y = x^2 - 18x + 10$.

- **a** Is the graph of the function concave up or down?
- **b** State the value of x at which the minimum value of the function occurs.
- **c** State the minimum y-value of the function.

5 Consider the quadratic function $y = x^2 - 6x + 5$.

- **a** Is the graph of the function concave up or down?
- **b** Find the x-intercepts.
- **c** State the the y-intercept.
- **d** Determine the axis of symmetry.
- **e** Hence or otherwise state the coordinates of the vertex of the parabola.

6 Consider the quadratic function $y = -x^2 - 2x + 8$.

- **a** Is the graph of the function concave up or down?
- **b** Find the x-intercepts of the function.
- **c** Find the y-intercept.
- **d** Determine the axis of symmetry.
- **e** Hence or otherwise find the vertex of the curve.

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7 Consider the quadratic function $y = -8 - 6x - x^2$.

a Is the graph of the function concave up or down?

b Find the x-intercepts of the function.

c Find the y-intercept.

d Determine the axis of symmetry.

e Hence or otherwise find the vertex of the curve.

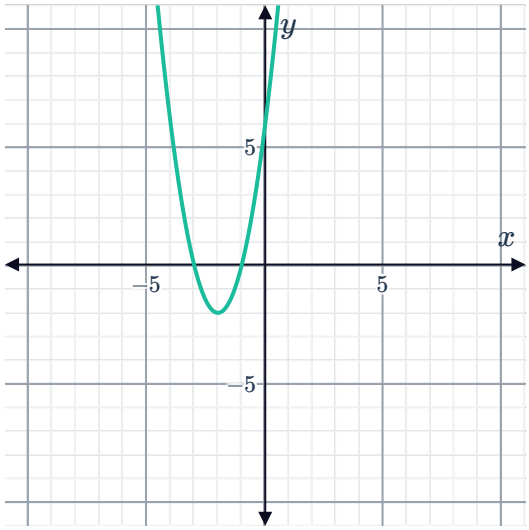
8 Consider the function $y = x^2 + 4x + 3$.

a Determine the equation of the axis of symmetry.

b Hence determine the minimum value of y.

c Hence state the coordinates of the vertex of the curve.

d Here is the graph of another quadratic function. State the coordinates of its vertex.



e Determine whether the following relationship is true between the graph of $y = x^2 + 4x + 3$ and the graph provided.

i They have the same x-intercepts.

ii They share the same turning point.

iii They have the same concavity.

9 Consider the function $y = (14 - x)(x - 6)$.

a State the zeros of the function.

b Find the axis of symmetry.

c Is the graph of the function concave up or concave down?

d Determine the maximum y-value of the function.

10 Find the coordinates of the vertex of $y = 3x^2 - 6x - 9$.

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11 Find the maximum value of y for the quadratic function $y = -x^2 + 10x - 25$.

12 Consider the quadratic function $f(x) = -7.5x^2 - 1.3x - 1.3$.

a Find the x-coordinate of the vertex correct to three decimal places.

b Hence, find the maximum value obtained by the function, correct to two decimal places.

13 Find the x-coordinate of the vertex of the parabola represented by $P(x) = px^2 - \frac{1}{2}px - q$.

14 Consider these two parabolas, labeled P_1 and P_2.
$P_1 : y = x^2 + 4x + 6$
$P_2 : y = x^2 - 4x + 6$

a Determine the coordinates of the vertex for each parabola.

b How far apart are the vertices of the two parabolas?

15 Consider the curve $y = x^2 + 6x + 4$.

a Determine the axis of symmetry.

b Hence, determine the minimum value of y.

Applications

16 When an object is thrown into the air, its height above the ground is given by the equation

$$h = 193 + 24s - s^2$$

where s is its horizontal distance from where it was thrown.

a Find the value of s, at the point when it reaches its greatest height above the ground.

b Find the maximum height reached by the object.

17 The height at time t of a ball thrown upwards is given by the equation $h = 59 + 42t - 7t^2$.

a How long does it take the ball to reach its maximum height?

b Find the height of the ball at its highest point.

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